

Logarithms and Their Properties

ANSWER KEY



Evaluating logarithms

1 Evaluate each logarithm exactly:

a $\log_2 32 = 5$

b $\log_3 \frac{1}{9} = -2$

c $\log 1000 = 3$

d $\ln e^7 = 7$

2 Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

$$\log_5 18 = \frac{\ln 18}{\ln 5} \approx \frac{2.8904}{1.6094} \approx 1.796.$$

3 State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

Domain of f : $x > 4$. Inverse of g : $g^{-1}(x) = \log_2 x$.

Properties of logarithms

4 Expand fully using properties of logarithms: $\ln \left(\frac{x^3 \sqrt{y}}{z^2} \right)$.
 $3 \ln x + \frac{1}{2} \ln y - 2 \ln z$.

5 Write as a single logarithm: $2 \log x + 3 \log y - \log z$.
 $\log \left(\frac{x^2 y^3}{z} \right)$.

6 Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6 = 1.114$

b $\log_b 8 = 1.293$

c $\log_b 1.5 = 0.252$